

# CSE 642: Scribe Notes

September 5, 2025

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## 1 Introduction

Quick discussion of problems to discuss.

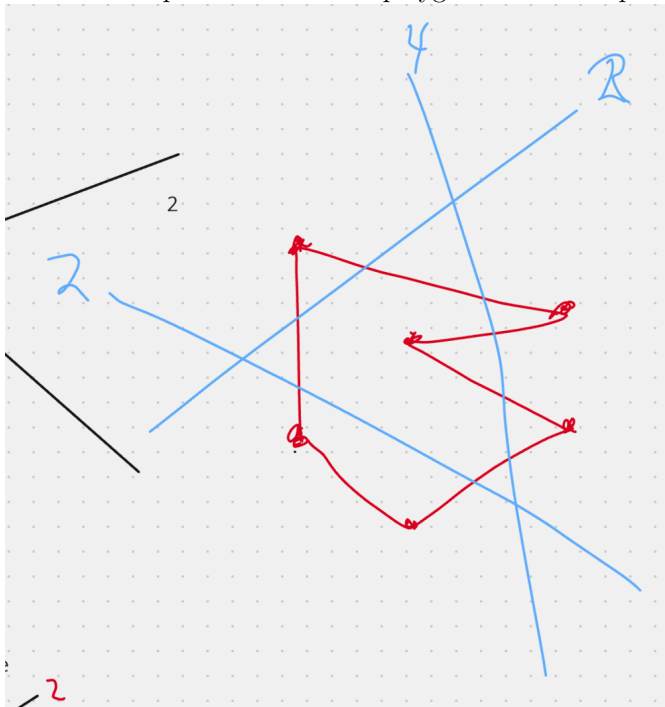
- We could resume conversation on polygon crosssections.
- Rezaul has some problems that are ready to discuss.
- Joe had a problem or two that are ready, including “Connections.”

## 2 Lucas’s Polygon Problem

### 2.1 Review the Problem Statement

Given a set  $S$  of  $n$  points on the plane, and a set  $L$  of  $m$  lines in the plane, all in general positions (no accidental points-on-lines, three points are co-linear, ect.). Each line has an integer label that is non-negative, and presumably even (we’re creating closed polygons). Question one: Does there exist a simple polygon  $P$  whose vertices are  $S$  and that cross each line the number of times given by its label. Question two: What is the complexity of deciding this and constructing the polygon if it exists.

Below is an example of a problem with a solution to question one. There are six vertices and three lines with labels. We see its possible to draw a polygon with the required crossings.



## 2.2 Review of previous discussion

We are searching for an algorithm that can answer this questions, and presumably return a correct construction. Last time we discussed some variations of this problem.

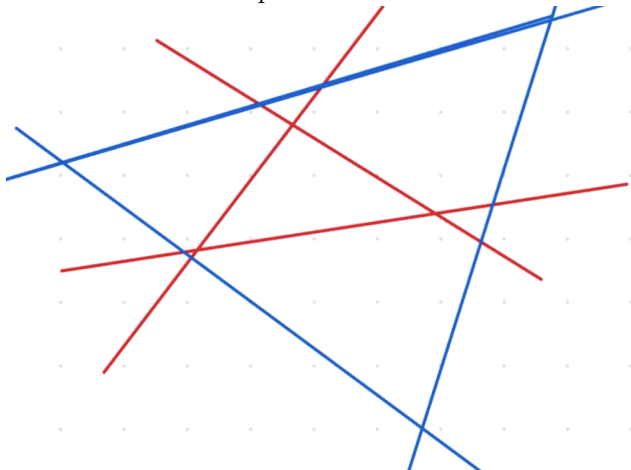
### 2.2.1 Introduce New Points

Can we introduce *Steiner points*, i.e. add extra points to the  $S$  to create a construction. In particular, if we just are given  $L$ , can we generate a set of points for a polygon that would create satisfy the crossing requirements?

### 2.2.2 Feasibility

Looked at example of all zero lines. Then all of  $S$  would have to lie within a given cell as defined by the line intersections.

We also looked at this example where red lines have label 0, and blue lines have level 2, which would not be feasible even with Steiner points.



## 2.3 New Discussion

We have not yet found the complexity.

Lucas found a Skienna paper that discusses a query version of this problem “[Reconstructing graphs from cut-set sizes.](#)” Here we have an unknown graph  $G$  with  $n$  vertices and straight edges, points in general position. We are given the vertex locations as input. In this setting we can ask cut queries, specified by a line  $L$  in the plane and returns how many edges are crossed. The Goal is to obtain a strategy to discover  $G$  completely using as few cuts as possible. In the paper, a general case algorithm is shown to require  $\mathcal{O}(n^2)$  queries are sufficient and sometimes necessary in general. The idea is that we can simulate an edge query with four cut queries. See the figure below.

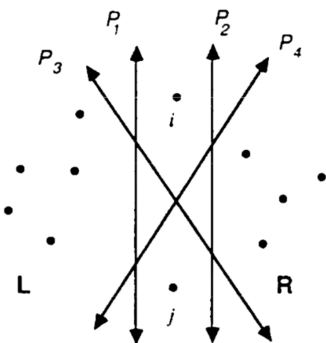


Fig. 1. The probes necessary to determine whether  $\{i, j\} \in E$ .

This was a problem that was discussed in this seminar in the Fall of 21 or 22. In the problem we’re currently discussing, we are given the queries and their answers. We then have to decide if there exists a cycle graph  $G$  on  $S$  with no crossing edges.

Question was asked: Are positive answers unique? Well trivially, if no lines are given, or all lines are zero and all points  $S$  lie in a common face, with the points not in convex position, then there are multiple polygons satisfying the constraints.

Another way to connect these two problems is to decide if the oracle is lying or not. That is, a motivation for Lucas's problem is to catch a lying oracle.

So does knowing something about the topology of what we're looking for (simple polygon, embedded tree, etc) help at all? This was an open question from the paper. We should look up scribe notes from when this was discussed in the seminar last time (Early October 2021?). There are other papers and results for probing graphs with cut queries. The main difference here is the geometry constraints.

Rezaul brought up a question: Given 2 polygons  $P, Q$  can they be translated / rotated so that they have  $k$  crossings? For pure translation, standard 2D computational geometry methods apply, and this would become a good homework problem. With rotations (possibly scaling, etc) more general methods apply. In particular, arrangements of lines and configuration spaces.

There is a related graph drawing problem: Given 2 fixed graphs  $G_1$  and  $G_2$ , translate / rotate / scale (with limitations) one with respect to the other in order to minimize (or max?) the number of crossings. Typically this would be accompanied by a bounded workspace, and limitations to prevent over scaling or translation for example.

## 2.4 Back on topic

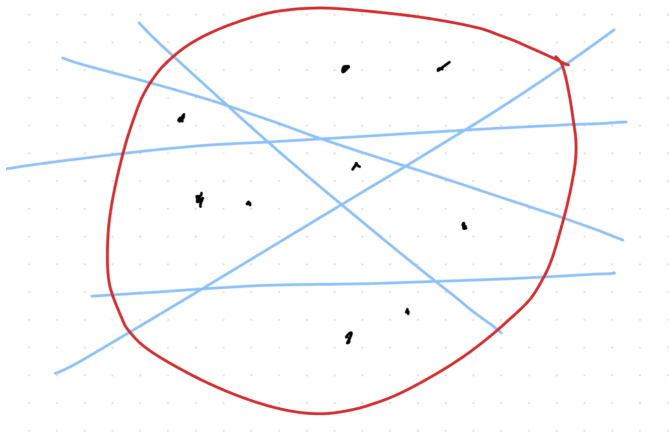
Last time we found that steiner points made the problem easier, with an initially empty  $S$ , and that its probably easy with non-empty  $S$ .

Some claims about easy cases.

Claim 1: It suffices to have labels of 0 or 2 on the lines (red / blue).

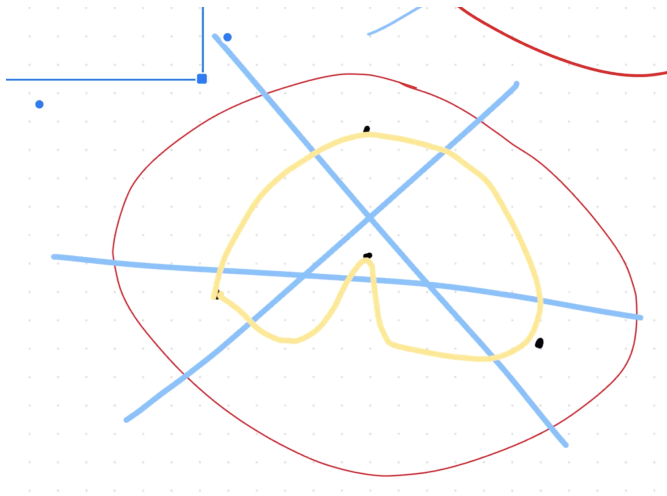
Claim 2: If there is a not a red line cell that is intersected by all blue lines, then it is hopeless, no polygon exists (no matter the set  $S$ ). The intuition here is that any simple polygon lives completely in a single red cell. And this can be checked in poly time.

So we now have our reduced easy case. Consider the case with a red line cell  $C$ , and all blue lines intersect this cell, and all the points  $S$  are contained in  $C$ . Consider the diagram below.



So can we, by adding additional points, can we draw a polygon through  $S$  that respects the blue crossing number. The idea here is that each blue line chops the problem in two, though there is some additional work to clarify the crossing number of problems within the sub-problem.

We can easily construct a problem example where this doesn't work, and at least one line must be crossed four times, see below. But again, can we decide this in polynomial time?



We can view this with a dual problem, looking at faces that must be visited.

We also discussed this in terms of a convex stabber, i.e. if we have to take a detour than at least on edge must be crossed four times.

Q: Must  $S$  be in convex position? (for there to be a solution)  
 ie, there is a solution iff there is a convex stabber of the occupied faces (those with points of  $S$ )

